C# COMPLETE TOPICS LIST

1 C# FUNDAMENTALS (ABSOLUTE BASICS)

Language Basics

- What is **C#**
- .NET Framework vs .NET Core vs .NET (modern)
- CLR (Common Language Runtime)
- CTS (Common Type System)
- CLS (Common Language Specification)
- JIT Compilation
- IL (Intermediate Language)

Program Structure

- `Main()` method
- Namespaces
- `using` keyword
- Console applications
- Project structure

Data Types

- Value types
- Reference types
- Built-in types:
  - `int`, `float`, `double`
  - `decimal`
  - `char`, `string`
  - `bool`
- `var`
- `dynamic`
- Nullable types (`int?`)

Variables & Constants

- Local variables
- Fields
- `const`
- `readonly`
- Scope and lifetime

2 OPERATORS & CONTROL FLOW

Operators

- Arithmetic operators
- Relational operators
- Logical operators
- Assignment operators
- Bitwise operators
- Ternary operator (`?:`)
- Null-coalescing (`??`)
- Null-conditional (`?.`)
- `is`, `as`

Control Statements

- `if`, `else`
- `switch`
- `for`
- `while`
- `do-while`
- `foreach`
- `break`, `continue`, `goto`
- Pattern matching in `switch`

3 METHODS & FUNCTIONS

- Method declaration

- Method overloading
- Optional parameters
- Named parameters
- `ref`, `out`, `in`
- Return types
- Expression-bodied methods
- Local functions

4️⃣ OBJECT-ORIENTED PROGRAMMING (CORE)

Classes & Objects

- Class definition
- Object creation
- Fields vs Properties
- Constructors
- Constructor overloading
- Static members
- Destructors / Finalizers

OOP Principles

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Access Modifiers

- `public`
- `private`
- `protected`
- `internal`
- `protected internal`
- `private protected`

5 PROPERTIES & STRUCTURES

Properties

- Auto-implemented properties
- Read-only properties
- Computed properties
- Property validation

Structs & Enums

- `struct`
- Value-type behavior
- `enum`
- Flags enum

6 INTERFACES & ABSTRACT CLASSES

Interfaces

- Interface definition
- Multiple inheritance via interfaces
- Default interface methods
- Interface segregation

Abstract Classes

- Abstract methods
- Abstract vs interface
- Partial implementation

7 COLLECTIONS & DATA STRUCTURES

Arrays

- Single-dimensional

- Multi-dimensional
- Jagged arrays

Collections

- `List<T>`
- `Dictionary<TKey, TValue>`
- `Queue<T>`
- `Stack<T>`
- `HashSet<T>`
- `SortedList`
- `ObservableCollection`

Interfaces

- `IEnumerable`
- `ICollection`
- `IList`
- `IDictionary`

8 EXCEPTION HANDLING

- `try-catch-finally`
- Multiple catch blocks
- Custom exceptions
- `throw` vs `throw ex`
- Exception hierarchy
- Best practices

9 FILE HANDLING & IO

- File vs Directory
- `File`, `FileInfo`
- `Directory`, `DirectoryInfo`
- `Stream`
- `FileStream`
- `StreamReader`, `StreamWriter`

- Serialization basics

10 DELEGATES, EVENTS & LAMBDAS

Delegates

- Delegate declaration
- Multicast delegates
- `Func<>`, `Action<>`, `Predicate<>`

Events

- Event pattern
- Event handlers
- Publisher-subscriber model

Lambda Expressions

- Lambda syntax
- Expression-bodied lambdas
- Usage with LINQ

11 LINQ (VERY IMPORTANT)

- What is LINQ
- LINQ to Objects
- LINQ to SQL / EF
- Deferred execution
- Immediate execution

LINQ Keywords

- `Where`
- `Select`
- `OrderBy`
- `GroupBy`
- `Join`

- `Any`, `All`
- `First`, `FirstOrDefault`
- `Single`, `SingleOrDefault`
- `Aggregate`

1️⃣2️⃣ GENERICS

- Generic classes
- Generic methods
- Constraints (`where T`)
- Covariance & Contravariance
- Performance benefits

1️⃣3️⃣ MEMORY MANAGEMENT

- Stack vs Heap
- Value type vs Reference type
- Garbage Collector (GC)
- Generations (Gen 0, 1, 2)
- `IDisposable`
- `using` statement
- Memory leaks

1️⃣4️⃣ MULTITHREADING & ASYNC

Threading

- `Thread`
- `Task`
- ThreadPool
- Synchronization
- `lock`
- `Monitor`
- `Mutex`

- `Semaphore`

Async Programming

- `async` / `await`
- Task-based async pattern
- Parallel programming
- Deadlocks
- Race conditions

1 5 ADVANCED C# FEATURES

- Extension methods
- Anonymous types
- Partial classes
- Records
- Tuples
- Pattern matching
- Span & Memory
- Unsafe code

1 6 REFLECTION & METADATA

- Reflection basics
- Type
- Assemblies
- Attributes
- Custom attributes
- Runtime type inspection

1 7 DEPENDENCY INJECTION & DESIGN

- Dependency Injection (DI)
- IoC containers

- SOLID principles
- Design patterns:
  - Singleton
  - Factory
  - Repository
  - Strategy
  - Observer
  - Adapter

1 8 DATABASE & DATA ACCESS

- ADO.NET
- Entity Framework
- Code First / Database First
- LINQ to Entities
- Transactions
- Connection pooling

1 9 TESTING & DEBUGGING

- Unit testing
- MSTest / NUnit / xUnit
- Mocking
- Debugging tools
- Logging
- Exception logging

2 0 APPLICATION TYPES

- Console applications
- Windows Forms
- WPF
- ASP.NET MVC
- ASP.NET Core

- Web APIs
- Microservices